



Lecture26_
MA5104

MA 5104: Statistical Inference and Learning

Module 4: Learning
Lecture 26: Linear Regression as Prediction

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Today

- 1 The Oldest Restriction
- 2 Least Squares
- 3 What a Coefficient Means
- 4 Categorical Predictors
- 5 Linear Smoothers
- 6 Closing

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The Oldest Restriction

Where the Name Comes From

Galton, 1886: children's heights against mid-parent heights.

Tall parents have tall children — but on average **less tall than the parents**.
Short parents, likewise, have children nearer the middle. He called it
regression toward mediocrity.

The word we use every day for fitting a line is a description of a biological
phenomenon that nobody now means when they say it.

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What We Are Actually Doing

Choose a loss, choose a function class, minimize empirical risk. Today we take

$$\mathcal{F} = \left\{ f(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p \right\}, \quad L(y, a) = (y - a)^2$$

- This is also the **rigid** end of the spectrum — high bias, low variance, and it escapes the curse of dimensionality because the restriction is strong.
- The whole question is what that restriction buys and what it costs.

We are doing prediction, not inference. No t -tests, no p -values, no significance. Those belong to the first half of this course and to your regression course — and they answer a different question.

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The Criterion and the Normal Equations

Stack the data: $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times (p+1)}$ with a leading column of ones. Minimize

$$\text{RSS}(\beta) = \|y - X\beta\|^2 = (y - X\beta)^T (y - X\beta)$$

Differentiate:

$$\frac{\partial \text{RSS}}{\partial \beta} = -2X^T(y - X\beta) = 0 \quad \implies \quad X^T X \hat{\beta} = X^T y$$

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If X has full column rank, $X^T X$ is invertible and

$$\hat{\beta} = (X^T X)^{-1} X^T y, \quad \hat{y} = X \hat{\beta} = \underbrace{X(X^T X)^{-1} X^T}_H y$$

Remember the form $\hat{y} = Hy$. We return to it at the end of the lecture, and it will turn out not to be special to this method.

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The Geometry: Least Squares Is a Projection

H is the orthogonal projection ^{operator} onto the column space of X .

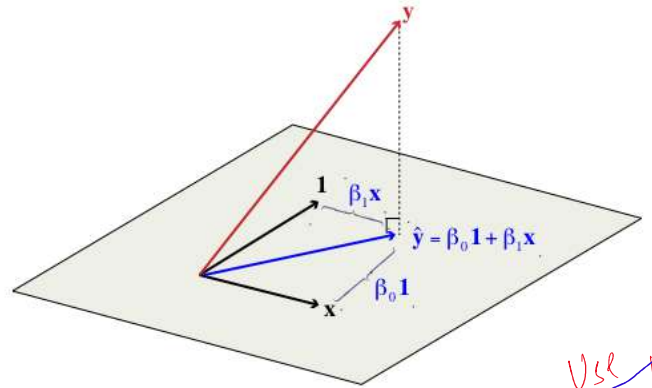
$$H = H^T, \quad H^2 = H, \quad X^T(y - \hat{y}) = 0$$

- $\hat{y}$ is the point in the column space **closest to y** .
- The residual $y - \hat{y}$ is orthogonal to **every** column of X — including the column of ones, so residuals sum to zero.
- $\text{tr}(H) = p + 1$: the projection uses up $p + 1$ dimensions of the n available.

Why the residual must be perpendicular

If it were not, you could slide $\hat{y}$ along the plane and get closer to y . **That is the entire proof**, and it needs no calculus.

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When It Fails: $p > n$

If X does not have full column rank — in particular whenever $p > n$ — then $X^T X$ is singular.

- $\hat{\beta}$ is not unique. Infinitely many coefficient vectors give **exactly** the same fitted values.
- In fact the fit is perfect: residuals are zero, training error is zero, and by Lecture 2 you already know what that is worth.
- Least squares simply has nothing to say here.

Modern data are routinely $p > n$: genomics, text, images. The fix is to shrink the class further — **that is Lecture 3** and it is why ridge and lasso exist at all.

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Marginal versus Partial

In simple regression, $\hat{\beta}_1$ describes the relationship between x_1 and y , **ignoring everything else**.

In multiple regression, $\hat{\beta}_1$ describes the relationship between x_1 and y **holding the other predictors fixed** — a partial effect.

These are different quantities

They answer different questions, and they can disagree in magnitude and **in sign**. This is not a pathology or a numerical problem. It is what the two coefficients mean.

Let us break it in front of you.

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if $\text{rank}(X^T X) < p$, then it is not good. cannot use LS $\Rightarrow$ Use lasso/ridge

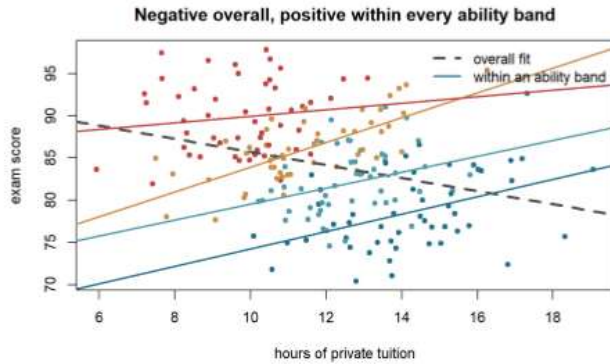
if $\text{rank}(X^T X) = p$, then you can use LS

$\leftrightarrow$ heights in cm = $f(\text{heights in inches})$

$\leftrightarrow \begin{bmatrix} X_1 \\ - \end{bmatrix} = f(X_2, X_5, X_8) + \epsilon$

$V(\hat{\beta})$ inflated
 $\hat{\beta}$ will not be precise
 Multicollinearity
 next course.

An example



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A Sign Flip, on Real Fitted Numbers

$n = 200$ students. Predict final exam score from hours of private tuition.

Model	coefficient on tuition	on cgpa
score ~ tuition	-0.90	—

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A Sign Flip, on Real Fitted Numbers

$n = 200$ students. Predict final exam score from hours of private tuition.

Model	coefficient on tuition	on cgpa
score ~ tuition	-0.90	—
score ~ tuition + cgpa	+1.46	+5.70

In these data $\text{corr}(\text{tuition}, \text{cgpa}) = -0.75$: the weaker students are the ones taking tuition.

Same data, same variable, same software. The sign changed because the question changed — from *what do tuition-takers score* to *what happens to a given student who takes more tuition*.

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Collinearity: Unstable Coefficients, Stable Predictions

When columns of X are nearly linearly dependent, $X^T X$ is nearly singular. Then:

What breaks

- $\hat{\beta}$ has huge variance
- coefficients swing wildly with small data changes
- individual coefficients become close to meaningless

What does not break

- $\hat{y}$ is stable
- prediction at new x in the same region is fine
- the *combination* is well determined even when the parts are not

This is the sharpest possible illustration of Lecture 1's split: a model can be excellent for **prediction** and useless for **interpretation**, simultaneously, on the same data. Lectures 7 and 8 do something about it.

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Dummy Coding

A predictor with k levels enters as $k - 1$ indicator columns, with one level held out as the **reference**.

Three departments, reference = Maths:

$$\hat{y} = \beta_0 + \beta_1 \mathbf{1}\{\text{Physics}\} + \beta_2 \mathbf{1}\{\text{Chemistry}\} + \dots$$

so β_1 is the Physics–Maths difference, not a Physics “effect”.

- Why $k - 1$ and not k : with all k columns plus an intercept, the columns sum to the intercept column. **X loses rank** and we are back on the previous failure slide.
- Changing the reference level changes every coefficient and **changes no fitted value whatsoever**.
- Ordinal predictors: coding as 1, 2, 3 asserts equal spacing. That is an assumption, and usually a wrong one.

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The Two Methods Have the Same Algebra

Least squares. $\hat{y} = Hy$ with $H = X(X^T X)^{-1} X^T$.

kNN. The fitted value at x_i is $\hat{y}_i = \frac{1}{k} \sum_{\ell \in N_k(x_i)} y_\ell$, which is $\hat{y} = Sy$ with

$$S_{i\ell} = \begin{cases} 1/k & \text{if } x_\ell \text{ is one of the } k \text{ neighbours of } x_i \\ 0 & \text{otherwise} \end{cases}$$

Both are linear smoothers

$\hat{y} = Sy$, where S depends on the x 's **but not on y** . Different shape of S — dense and global for least squares, sparse and local for kNN — but the same object.

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Four Things to Take Away

- ❶ Least squares is orthogonal projection onto the column space of X . The geometry is what to remember; the algebra follows from it.
- ❷ A coefficient is a partial effect, and it can flip sign relative to its marginal version. There is no such thing as *the* effect of a variable — only its effect given what else is in the model.
- ❸ Collinearity destabilizes coefficients while leaving predictions intact. A model can be excellent for prediction and useless for interpretation at the same time.
- ❹ Least squares and k NN are both linear smoothers